Hepatic reserve, portal failure, biliary sepsis, pancreas

Three axes of liver tests

- Injury, bile flow, functional reserve
- Aminotransferases leak from injured cells
  - Fall with recovery or with few hepatocytes left
- Prothrombin time fast, albumin slow
- Pattern: aminotransferase versus phosphatase ratio
  - Hepatocellular: viral, toxic, immune, ischaemic
  - Cholestatic: obstruction, infiltration, small ducts
- Modest values can accompany advanced cirrhosis

Jaundice and drug-induced injury

- Localise jaundice before imaging
- Conjugated bilirubin: excretory failure or obstruction
- Painful febrile jaundice: infected obstruction
- Painless jaundice: pancreatic, biliary, hepatic cancer
- Complete exposure timeline with re-exposure
- Predictable dose-related versus idiosyncratic delayed
- Rechallenge can provoke more severe injury

Cirrhotic circulation and haemostasis

- Septa and nodules raise sinusoidal resistance
- Splanchnic vasodilation lowers effective volume
  - Kidneys perceive underfilling, retain sodium
- Collaterals bypass clearance, form varices
- Splenic congestion lowers platelets
- Fragile rebalanced haemostasis
  - Tests miss loss of anticoagulant factors

Variceal bleeding

- Avoid excess volume that raises portal pressure
- Protect airway, vasoactives, antibiotics, ligation
- Tamponade is a temporary bridge
- Early shunt helps high risk, worsens encephalopathy
- Secondary prevention: beta blocker plus ligation
- Look for coexisting lesions such as ulcer

Acute pancreatitis

- Enzyme activation, then vascular leak
- Compatible combination, not all three findings
- Enzyme height does not grade severity
- Persistent organ failure predicts severe disease
- Titrated crystalloid, not aggressive fluid
  - Excess worsens lung and abdominal oedema
- Early enteral nutrition preserves gut integrity
- Delay intervention, step-up approach
- Urgent cholangiopancreatography: cholangitis, obstruction

Acute liver failure

- Danger extends beyond bleeding
  - Cerebral oedema, hypoglycaemia, shock, kidney injury
- Serial glucose, neurology, acid-base in intensive care
- Sedatives cloud encephalopathy assessment
- Prompt N-acetylcysteine if paracetamol suspected
  - Unknown timing does not exclude paracetamol
- Seek causes in parallel, do not delay referral
- Prophylactic plasma obscures risk and transplant decision

Ascites and peritonitis

- Sample when new, worsening, or admitted
- Blood-culture bottles improve culture yield
- Albumin gradient estimates portal cause
- Normal white count or no fever does not exclude
- Routine coagulopathy does not delay diagnosis
- Treat at neutrophil threshold before cultures
- Polymicrobial or localised signs: secondary peritonitis
- Diuretics balance retention against effective volume
  - Overdiuresis: kidney injury, hyponatraemia
- Large-volume drainage needs albumin replacement

Encephalopathy and kidney injury

- Ammonia correlates poorly with the syndrome
- Search bleeding, infection, diuresis, sedatives
- Lactulose traps nitrogen in the colon
- Kidney injury is not automatically hepatorenal
  - Volume loss, sepsis, nephrotoxins assessed first
- Hepatorenal: vasodilation plus renal vasoconstriction
  - Albumin challenge supports diagnosis
- Urinary sodium alone misleads
- Monitor ischaemia during vasoconstrictor therapy

Steatotic and biliary disease

- Steatotic disease part of insulin resistance
  - Fibrosis stage outweighs steatosis
- Non-invasive scores avoid biopsying everyone
- Biliary pain steady, not truly colicky
- Cholangitis is sepsis behind an obstructed duct
  - Antibiotics cannot sterilise without drainage

Chronic pancreatitis, cancer, and synthesis

- Exocrine and endocrine destruction
  - Low glucagon reserve: hypoglycaemia risk
- Enzymes in lipase units from first mouthful
- Pain care beyond opioids alone
- Cancer presents through obstruction, not mass
  - Coordinate so procedures do not delay surgery
- Improving tests are not recovery
- Name owner and time of reassessment